DANIEL KARIMI

SOFTWARE ENGINEER · DATA PROCESSING

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Talent Attraction Team – Indra Space
Indra Space Business Unit
Madrid, Spain

RE: Application for Software Engineer (MetOp-SG Data Processing Algorithms)

Dear Talent Attraction Team at Indra Space,

I am writing to express my eager interest in the **Software Engineer** position within Indra Space, focusing on the evolution and maintenance of data processing algorithms for the MetOp-SG mission. As a soon-to-be graduate in **Telecommunications Systems Engineering from Universidad Politécnica de Madrid (UPM)**, my academic and hands-on professional experiences have centered heavily on bridging the gap between complex mathematical algorithms and robust, efficient software implementations in Java, Python, and C++.

During my engineering coursework and subsequent research internship at the **Spanish National Research Council (CSIC)**, I developed a deep competency in designing, optimizing, and deploying data-intensive processing applications. At the CSIC, I was responsible for taking theoretical algorithms—much like the scientific updates expected during sensor commissioning phases—and architecting them into scalable pipelines. By emphasizing rigorous **unit testing and profiling memory and CPU utilization**, I successfully reduced algorithm execution times by orders of magnitude.

Why my profile aligns with the role:
· Strong academic foundation in Java (concurrency, streams), C++, and GNU/Linux environments.
· Deep understanding of distributed systems and parallel processing paradigms (relevant to Apache Kafka/Storm integrations).
· Proven ability to maintain and optimize complex scientific data processing algorithms for high performance.
· Hands-on experience with unit testing, Maven build systems, Docker, and CI/CD pipelines.

A core component of my academic background involved studying distributed architectures and networking protocols. The coursework required implementing multi-threaded Java applications that emphasized resource synchronization and data serialization—the very paradigms that underpin distributed payload processing frameworks like Apache Kafka and Storm. Combined with my daily usage of **GNU/Linux environments and Bash scripting**, I am highly prepared to integrate these processing algorithms within Indra's mission-critical frameworks.

Furthermore, I understand the critical importance of producing high-quality technical documentation and engaging proactively with validation teams to ensure operational readiness. My recent collaborative work in scientific research environments has instilled in me a team-oriented mindset, an aptitude for fast learning, and an unwavering commitment to coding standards and software development best practices.

I am deeply inspired by Indra Space's mission to redefine the European space industry and its contributions to flagship programs like Copernicus and Galileo. I am eager to bring my strong engineering foundation and modern software practices to your team. Please find my resume and a recommendation letter attached for your review. I look forward to the opportunity to discuss my contributions to the MetOp-SG mission in an interview.

Sincerely,

DANIEL KARIMI ÁLVAREZ
BSC TELECOMMUNICATIONS SYSTEMS ENGINEERING · UPM